

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the Intelligent Flood Prediction and Early Warning System.

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Category	Priority
1	User Input	Accept weather data from the user.	High
2	Data Preprocessing	Clean and prepare the input data.	High
3	ML Model Integration	Connect the trained ML model with Flask.	High
4	Flood Prediction	Predict flood risk based on input data.	High
5	Result Display	Show prediction results on the web page.	Medium
6	Model Training	Train ML models using historical flood data.	High
7	Input Validation	Validate user inputs before prediction.	High
8	Model Storage	Load the trained model using Pickle.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance	Fast prediction response.	< 5 Seconds
2	Scalability	Handle large datasets.	10,000+ Records
3	Security	Validate user inputs.	Secure Input Validation
4	Reliability	Provide consistent predictions.	99% Availability
5	Usability	Simple and responsive interface.	User friendly
6	Maintainability	Easy to update and maintain.	Modular design